

Olist Brazilian E-Commerce Data Dictionary

This document serves as the official data dictionary for the Olist E-Commerce dataset, defining the tables, columns, primary keys, foreign keys, and descriptions.

1. Table Definitions

1.1 Customers Table (`olist\_customers\_dataset.csv`)

Contains customer location details. A customer has a unique `customer\_id` for every order, but their overall customer identity is captured by `customer\_unique\_id`.

Column	Physical Type	Key	Description
`customer_id`	VARCHAR	PK	Order-level customer key. Links to `FactOrders`.
`customer_unique_id`	VARCHAR	-	Unique customer identifier (remains constant across orders).
`customer_zip_code_prefix`	INTEGER	FK	First 5 digits of customer zip code. Links to `DimGeolocation`.
`customer_city`	VARCHAR	-	City name.
`customer_state`	VARCHAR	-	State code (e.g., SP, RJ).

1.2 Geolocation Table (`olist\_geolocation\_dataset.csv`)

Contains spatial coordinates mapping zip code prefixes to latitude/longitude centroids.

Column	Physical Type	Key	Description
`geolocation_zip_code_prefix`	INTEGER	PK/FK	First 5 digits of zip code.
`geolocation_lat`	DOUBLE	-	Latitude coordinate centroid.
`geolocation_lng`	DOUBLE	-	Longitude coordinate centroid.
`geolocation_city`	VARCHAR	-	Normalized city name.
`geolocation_state`	VARCHAR	-	Normalized state code.

1.3 Order Items Table (`olist\_order\_items\_dataset.csv`)

Tracks line-item details for each order transaction.

Column	Physical Type	Key	Description
`order_id`	VARCHAR	FK	Unique order identifier. Links to `FactOrders`.
`order_item_id`	INTEGER	PK	Sequential integer identifying items in the same order.
`product_id`	VARCHAR	FK	Product identifier. Links to `DimProducts`.
`seller_id`	VARCHAR	FK	Seller identifier. Links to `DimSellers`.
`shipping_limit_date`	TIMESTAMP	-	Date by which the seller must hand over the product.
`price`	DOUBLE	-	Unit selling price.
`freight_value`	DOUBLE	-	Shipping fee per unit.

1.4 Order Payments Table (`olist\_order\_payments\_dataset.csv`)

Tracks payment transaction details. Multiple rows can exist per order if split payments are used.

Column	Physical Type	Key	Description
`order_id`	VARCHAR	FK	Unique order identifier. Links to `FactOrders`.
`payment_sequential`	INTEGER	PK	Sequential count for split payment methods.
`payment_type`	VARCHAR	-	Method used (credit_card, debit_card, voucher).
`payment_installments`	INTEGER	-	Selected credit card installments count.
`payment_value`	DOUBLE	-	Total payment transaction value.

1.5 Order Reviews Table (`olist\_order\_reviews\_dataset.csv`)

Tracks customer review ratings and written feedback.

Column	Physical Type	Key	Description
`review_id`	VARCHAR	PK	Unique review identifier.
`order_id`	VARCHAR	FK	Unique order identifier. Links to `FactOrders`.
`review_score`	INTEGER	-	Rating from 1 (lowest) to 5 (highest).
`review_comment_title`	VARCHAR	-	Title of review text comment (defaulted to 'no_me
`review_comment_message`	VARCHAR	-	Written review message (defaulted to 'no_me
`review_creation_date`	TIMESTAMP	-	Date the survey was sent.
`review_answer_timestamp`	TIMESTAMP	-	Timestamp of customer submission.

1.6 Orders Table (`olist\_orders\_dataset.csv`)

The central transaction table tracking states and logistics timestamps.

Column	Physical Type	Key	Description
`order_id`	VARCHAR	PK	Unique order identifier.
`customer_id`	VARCHAR	FK	Order-level customer key. Links to `DimCustomer`.
`order_status`	VARCHAR	-	State of transaction (delivered, shipped, canceled).
`order_purchase_timestamp`	TIMESTAMP	-	Date and time order was placed.
`order_approved_at`	TIMESTAMP	-	Timestamp of payment approval.
`order_delivered_carrier_date`	TIMESTAMP	-	Timestamp carrier picked up package.
`order_delivered_customer_date`	TIMESTAMP	-	Timestamp package reached customer.
`order_estimated_delivery_date`	TIMESTAMP	-	Estimated delivery date promised by Olist.

1.7 Products Table (`olist\_products\_dataset.csv`)

Contains product descriptions and physical characteristics.

Column	Physical Type	Key	Description
`product_id`	VARCHAR	PK	Unique product identifier.
`product_category_name`	VARCHAR	FK	Category name in Portuguese. Links to translation table.

`product_category_name_english`	VARCHAR	-	Translated English category name (defaulted
`product_weight_g`	DOUBLE	-	Product weight in grams.
`product_length_cm`	DOUBLE	-	Product length in cm.
`product_height_cm`	DOUBLE	-	Product height in cm.
`product_width_cm`	DOUBLE	-	Product width in cm.

1.8 Sellers Table (`olist\_sellers\_dataset.csv`)

Contains seller geographic data.

Column	Physical Type	Key	Description
`seller_id`	VARCHAR	PK	Unique seller identifier.
`seller_zip_code_prefix`	INTEGER	FK	First 5 digits of seller zip code. Links to `DimG
`seller_city`	VARCHAR	-	Normalized seller city.
`seller_state`	VARCHAR	-	Seller state code.